PASTA worksheet

Stages	Sneaker company
I. Define business and security objectives	• *The app will process transactions and provide multiple payment options for users.* • *The app will store and manage user information, including account details and seller profiles.* • *The app must protect user data and comply with payment-related security requirements to avoid legal issues.*
II. Define the technical scope	*I would prioritize evaluating SQL because the application uses databases to store user and product information. Poorly secured SQL databases can be vulnerable to SQL injection attacks, which may allow attackers to access sensitive data. Protecting database queries and using secure coding practices can reduce these risks.*
III. Decompose application	Sample data flow diagram
IV. Threat analysis	• Internal threat: • An employee with unauthorized access could misuse sensitive customer information stored by the application. • External threat: • A threat actor could perform SQL injection attacks to access sensitive user and payment

	data.
V. Vulnerability analysis	• *1. SQL injection vulnerabilities caused by insecure code that does not properly validate user input.* • *2. Database security weaknesses that could expose sensitive user and payment information to unauthorized users.*
VI. Attack modeling	Sample attack tree diagram
VII. Risk analysis and impact	1. Multi-factor authentication (MFA) to prevent unauthorized access to user accounts. 2. Input validation and prepared statements to prevent SQL injection attacks. 3. Encryption of sensitive data using secure encryption methods to protect user and payment information. 4. Role-based access control (RBAC) to ensure users only have the permissions they need.